

# SILIANG LU

(+86)13818564073

lusiliang93@gmail.com

I am currently working as a senior expert in Bosch Corporate Research. I have been leading projects in industrial AI across domains with different business units, especially control & optimization, time-series analysis in mobility, energy & supply chains. With proven tracking records in leading cross-functional collaborations, I have delivered scalable AI solutions to BU customers for achieving business goals or cost reduction.

## EDUCATION

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| <b>Doctor of Philosophy in Building Performance and Diagnostics</b><br>Carnegie Mellon University, Pittsburgh, USA<br>Specialization: Reinforcement learning for building system control | <i>Aug. 2015 - Sep. 2019</i> |
| <b>Master of Science in Building Performance and Diagnostics</b><br>Carnegie Mellon University, Pittsburgh, USA                                                                          | <i>Aug. 2015 - May 2017</i>  |
| <b>Master of Business Administration (MBA)</b><br>Fudan University, Shanghai, China                                                                                                      | <i>Sep. 2021 - Dec. 2023</i> |
| <b>Bachelor of Science in Building Science</b><br>Donghua University, Shanghai, China                                                                                                    | <i>Sep. 2011 - Jun. 2015</i> |

## INDUSTRIAL EXPERIENCE

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| <b>Senior Expert</b><br>Bosch Corporate Research, Shanghai, China                       | <i>Jul. 2024 - present</i>   |
| <b>Senior AI Scientist</b><br>Bosch Corporate Research, Singapore                       | <i>Apr. 2023 - Sep. 2023</i> |
| <b>Senior AI Scientist</b><br>Bosch Corporate Research, Shanghai, China                 | <i>Jan. 2021 - Jun. 2024</i> |
| <b>AI Scientist</b><br>Bosch Center for Artificial Intelligence (BCAI), Shanghai, China | <i>Sep. 2019 - Dec. 2020</i> |

## SELECTED PROJECTS

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| <b>XYZ vehicle integrated control with reinforcement learning</b><br><i>Corporate Research, Bosch China</i> <ul style="list-style-type: none"><li>- Lead and collaborate with domain experts on the data-driven AI vehicle dynamics controller research with reinforcement learning and predictive safety filter in CarSim simulation &amp; double track model (differentiable 7DoF vehicle dynamics model) to reduce manual calibration effort and enhance comfort &amp; stability.</li></ul> | <i>Mar. 2025 - present</i> |
| <b>Data-driven energy system optimization</b><br><i>Corporate Research, Bosch China</i> <ul style="list-style-type: none"><li>- Lead the large-scale multi-chiller energy system optimization platform with data-driven cooling load forecast model and real-time control in over 8 plants for manufacturing plant energy savings with facility management teams. (Bosch global top inventor, 3 patents filed).</li></ul>                                                                      | <i>Jan. 2020 - present</i> |

- Lead the large-scale AHU & VAV energy system flexibility optimization with data-driven indoor temperature forecast model and model predictive control for manufacturing plant energy savings. (1 patent filed, 1 publication).

### **Hybrid capacitated truck planning for external milkrun**

*Mar.2023 - Jun.2024*

*Corporate Research, Bosch China*

- Lead the development of truck routing optimization algorithms to minimize transportation costs, including slack induction & string removal (SISR) and transformer-based reinforcement learning planner (2 patents filed).

### **Data-driven auto-calibration of chassis controllers**

*Apr.2024 - Dec.2024*

*Corporate Research, Bosch China*

- Lead the development of axle velocity forecast model as offline environment/simulator for multi-objective optimization of 12 different PID parameters with genetic optimization.
- Validated the performances of the calibrated parameters in real vehicle tests with different types and different road surfaces.

### **Time-series diesel pump predictive diagnostics**

*Jan.2022 - Dec.2024*

*Corporate Research, Bosch China*

- Lead the real-time anomaly detection function for pre-diagnostics with self-recovery embedded in the ECU of the high pressure diesel pump of truck braking system with current forecast model by MLP, auto-encoder, etc with real vehicle and testbench data so as to reduce ZME sticky effects.
- The solution has been in the software platform since 2024/Q4 for GWM to enhance pump quality in the market, enhance OEM's satisfaction (2 patents filed). [link](#)

## **SELECTED PUBLICATIONS**

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**Siliang Lu\***, Baocong Zhang\*, Chenyang Li (2026). Predictive safety filter enhanced curriculum learning control for efficient vehicle dynamics controller. arXiv.

**Siliang Lu**, Dan Hu, Lili Wu (2026). Vehicle routing problem using deep reinforcement learning – A case study about truck planning in the industry. arXiv.

Liping Sun, Yucheng Guo, **Siliang Lu**, Zhenzhen Li. Time-series Forecast for Indoor Zone Air Temperature with Long Horizons A Case Study with Sensor-based Data from a Smart Building. ICML Co-build workshop (oral). Vancouver, Canada. Jul., 2025.

**Siliang Lu**, Pengkun Liu. Real-Time Chiller Optimization in Industrial Plants based on Data-Driven Load Forecast with domain knowledge. ASCE International conference on Computing in Civil Engineering. ASCE. Pittsburgh, USA. 2024.

**Siliang Lu**, Erica Cochran Hameen. An Interactive Task Conditioning System Featuring Personal Comfort Models and Non-Intrusive Sensing Techniques: A field Study in Shanghai. Technologies [J]. 2021

**Siliang Lu**, Zhiang Zhang, Erica Cochran Hameen, Berangere Lartigue, Omer Karaguzel. Energy Co-Simulation of the Hybrid Cooling Control with Synthetic Thermal Preference Distributions. Symposium on simulation in architecture + urban design (SimAUD), ACM. Online Conference. 2020.

**Siliang Lu**, Weilong Wang, Shihan Wang, Erica Cochran Hameen. Thermal comfort-based personalized models with non-intrusive sensing technique in office buildings. Applied Sciences[J]. 2019.

**Siliang Lu**, Weilong Wang, Chaochao Lin, Erica Cochran Hameen. Data-driven simulation of a thermal comfort-based temperature set-point control with ASHRAE RP884. Building and environment [J].2019.

**Siliang Lu**, Erica Cochran Hameen. Simulation Study of Individual Thermal Comfort with the Integrated Personalized Fan and VAV System. IBPSA Building Simulation Conference, Rome, Italy. September, 2019.

**Siliang Lu**, Erica Cochran Hameen. Integrated infrared temperature sensor for online clothing insulation measurement. Computer-aided architectural design research in Asia. Beijing, China. May. 2018.

**Siliang Lu**, Erica Cochran Hameen, Azizan Aziz. Dynamic HVAC Operations with Real-Time Vision-Based Occupant Recognition System (Best student paper finalist). The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Winter conference, Chicago, USA. January, 2018.

## AWARDS

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2018 Young Computer-aided Architectural design research in Asia (CAADRIA) Award

2018 Honorable mentions in ASHRAE Best Student Paper, ASHRAE Winter Conference

2025 Changning Craftsman, Changning Labor Union, Shanghai

2022 Global Top Inventor (10 global inventors in total), Bosch Group [link](#)

2022 APEC sustainable production and consumption Award for Bosch [link](#)

2022 China commercial vehicle black technology award, China Association of Automobile Manufacturers

2021 Sustainable Business Awards of EU Chamber of Commerce in China for Bosch